FCC-ue & Muon-Collider $4b$ Production

01 Original charged-current study — COMPLETED

	FCC-ue: μ^+e^-	Muon collider: $\mu^+\mu^-$
A. Double-Higgs	$\mu^+e^- \to \nu_e\bar{\nu}_\mu HH$ $\to \nu_e\bar{\nu}_\mu\, b\bar{b}\, b\bar{b}$	$\mu^+\mu^- \to \nu_\ell\bar{\nu}_\ell HH$ $\to \nu_\ell\bar{\nu}_\ell\, b\bar{b}\, b\bar{b}$
B. Complete CC $4b$	$\mu^+e^- \to \nu_e\bar{\nu}_\mu\, b\bar{b}\, b\bar{b}$	$\mu^+\mu^- \to \nu_\ell\bar{\nu}_\ell\, b\bar{b}\, b\bar{b}$

02 Additional neutral-current study: FCC-ue, unpolarized 1 TeV — COMPLETED

NC double-Higgs signal	$\mu^+e^- \to \mu^+e^- HH \to \mu^+e^-\, b\bar{b}\, b\bar{b}$
NC associated Higgs	$\mu^+e^- \to \mu^+e^-\, b\bar{b}H \to \mu^+e^-\, b\bar{b}\, b\bar{b}$
NC $4b$ continuum	$\mu^+e^- \to \mu^+e^-\, b\bar{b}\, b\bar{b}$

03 Two requested polarized NC associated-Higgs extensions — Running

FCC-ue	$\mu^+e^- \to \mu^+e^-\, b\bar{b}H \to \mu^+e^-\, b\bar{b}\, b\bar{b}$
	$E_{\mu^+} = 5000$ GeV; $E_{e^-} = 183$ GeV; $P_{\mu^+} = +80\%$, $P_{e^-} = 0$.
Muon collider	$\mu^+\mu^- \to \mu^+\mu^-\, b\bar{b}H \to \mu^+\mu^-\, b\bar{b}\, b\bar{b}$
	$E_{\mu^+} = E_{\mu^-} = 1000$ GeV ($\sqrt{s} = 2$ TeV); $P_{\mu^+} = +80\%$, $P_{\mu^-} = 0$.